

Find $\rightarrow \infty$ || 5 attempts

25% HW $\times \frac{100}{80}$

20% T1 $\times \frac{100}{80}$

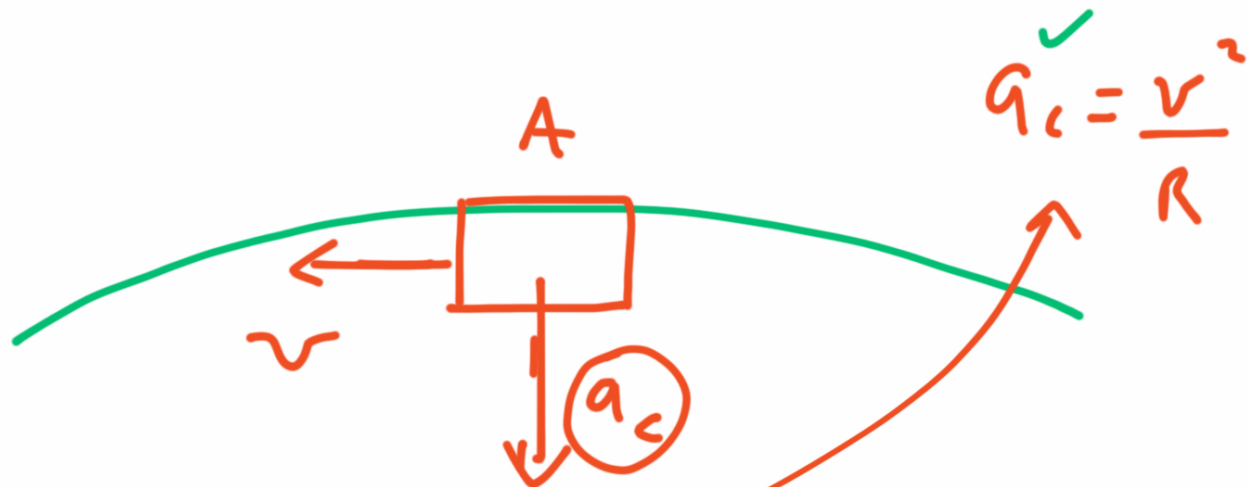
~~20% T2~~

$\hookrightarrow$ 35% Find $\frac{100}{80}$

80%

c) 5
b) 5
c) 5
d) 5

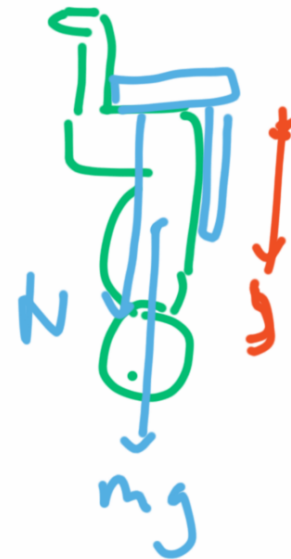


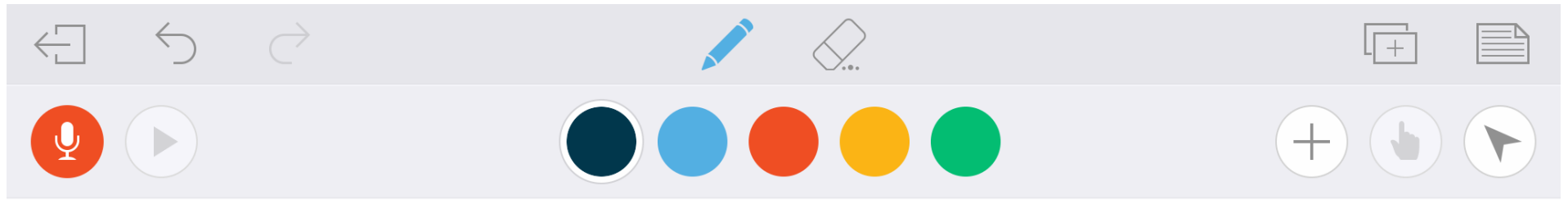


$$\sum \vec{F}_j = m \vec{a}_j$$

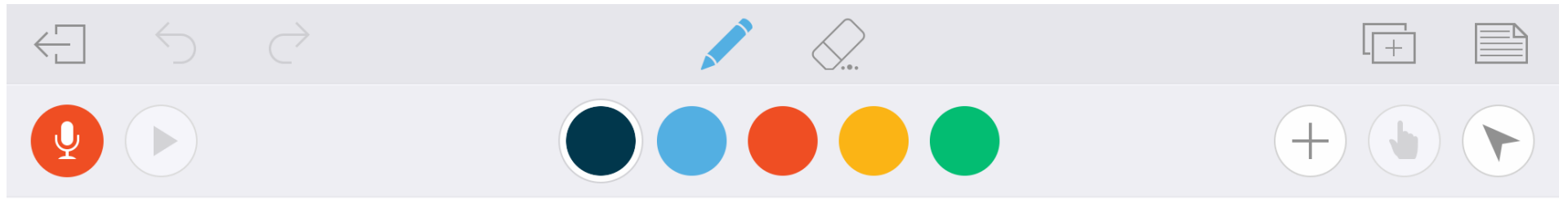
$$mg + N = m(a_c)$$

$$mg + N = m \frac{v^2}{R}$$





$$\begin{aligned} N &= m \frac{v^2}{R} - mg \\ &= m \left(\frac{v^2}{R} - g \right) \\ &= 40 \left(\frac{15.0^2}{8.6} - 9.8 \right) \\ &= 655 \text{ N} \end{aligned}$$



$$a_c = \frac{v^2}{R}$$

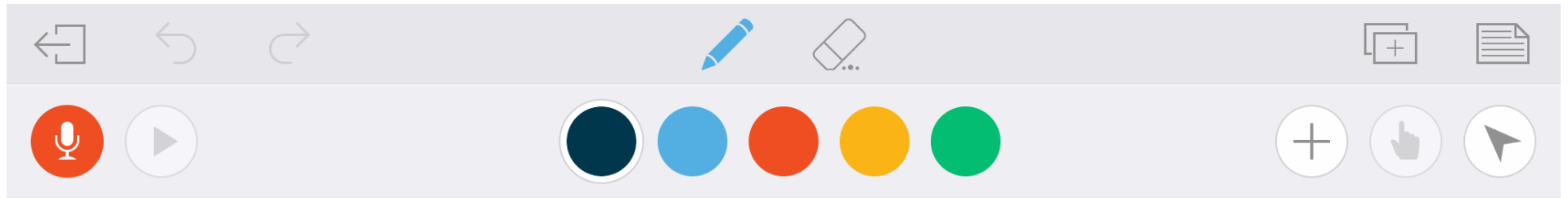


$$\Sigma \vec{F}_g = m \vec{a}_g$$

$$N + \underline{mg \sin \theta} = m \underline{\frac{v^2}{R}}$$

30°





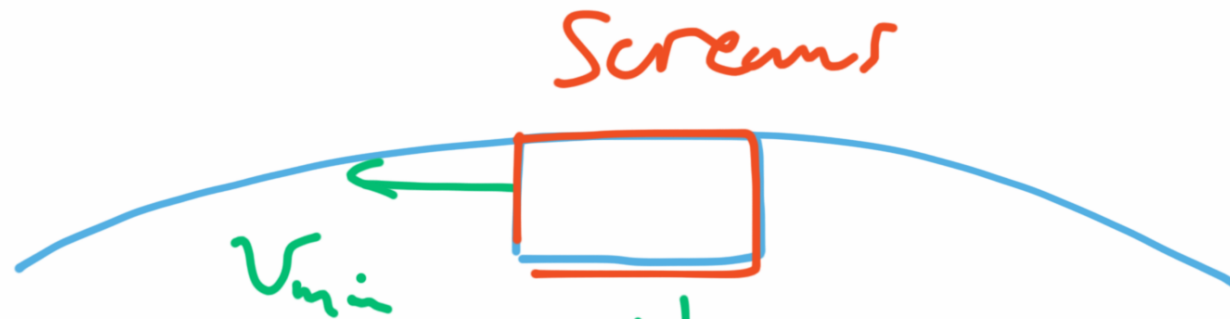
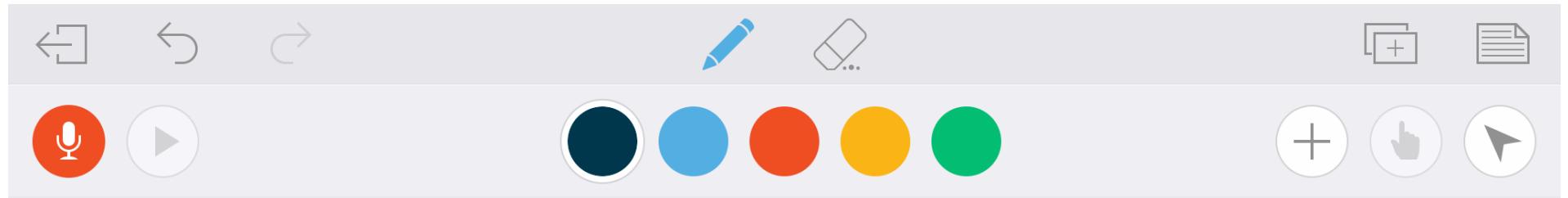
$$N = \frac{mv^2}{R} - mg \sin \theta$$

bigger

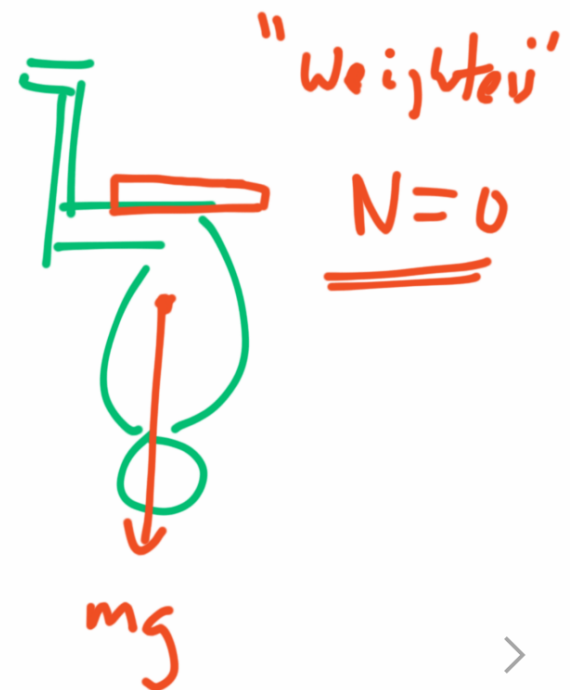
smaller

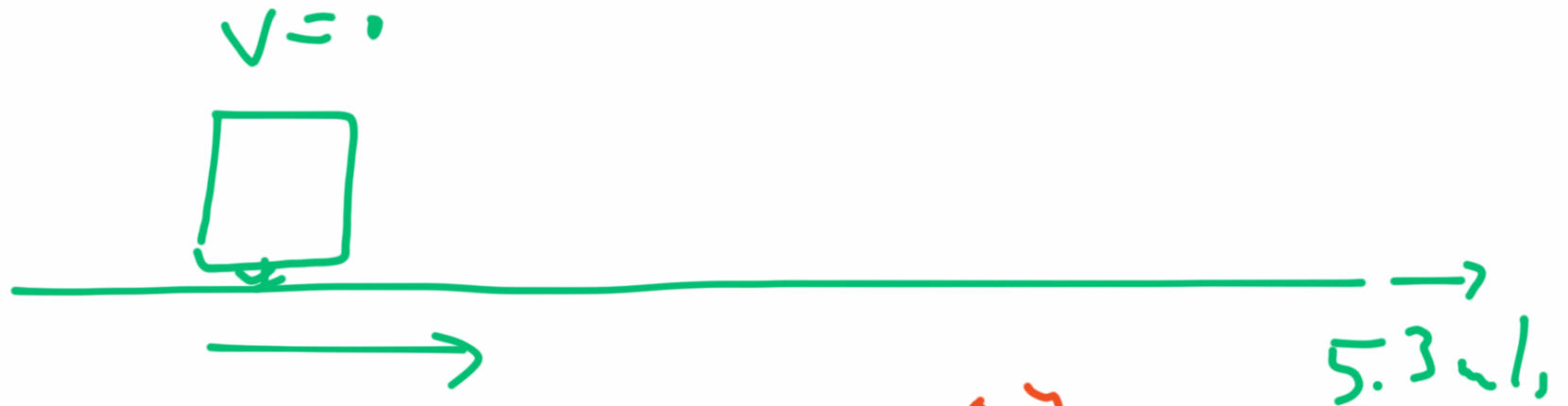
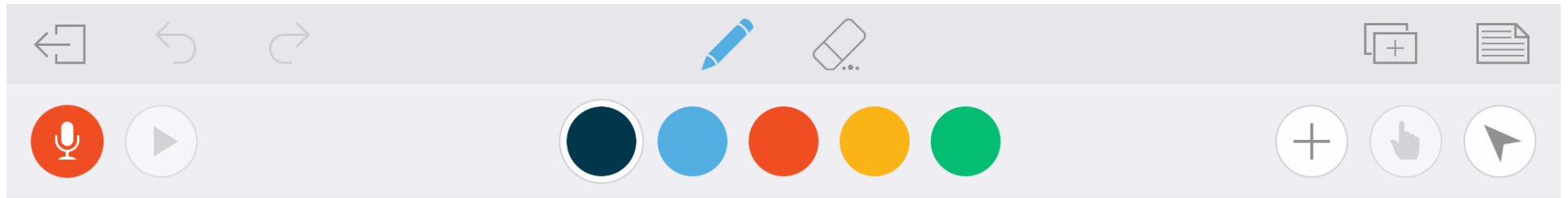
$$= \underline{807 \text{ N}}$$

$$4) N = \frac{mv^2}{R} - mg \sin \theta$$



$$\begin{aligned}\Sigma F_y &= ma_y \\ \cancel{mg} &= \cancel{m} \frac{v_{min}^2}{R} \\ v_{min} &= \sqrt{gR} \\ &= 9.18 \text{ m/s}\end{aligned}$$

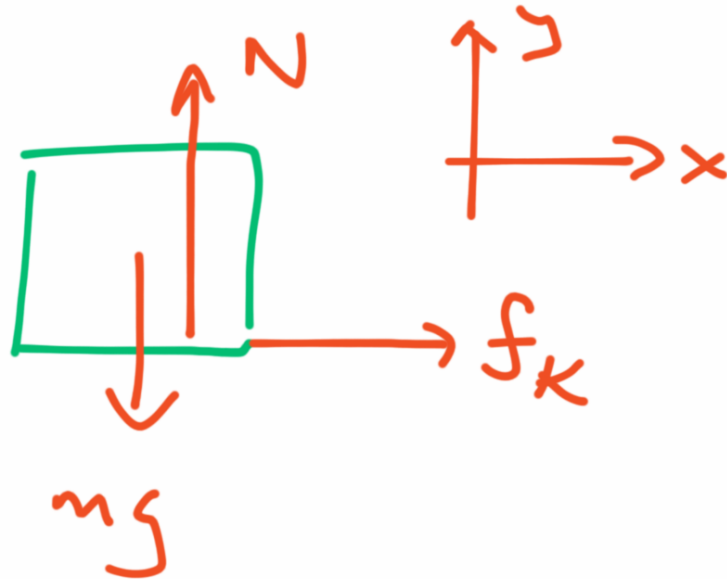


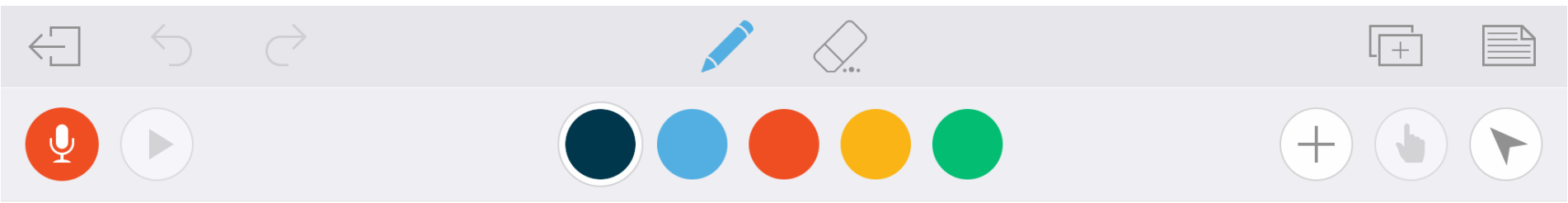


$$\sum \vec{F}_j = m \vec{a}_j$$

$$N - mg = 0$$

$$\boxed{N = mg}$$



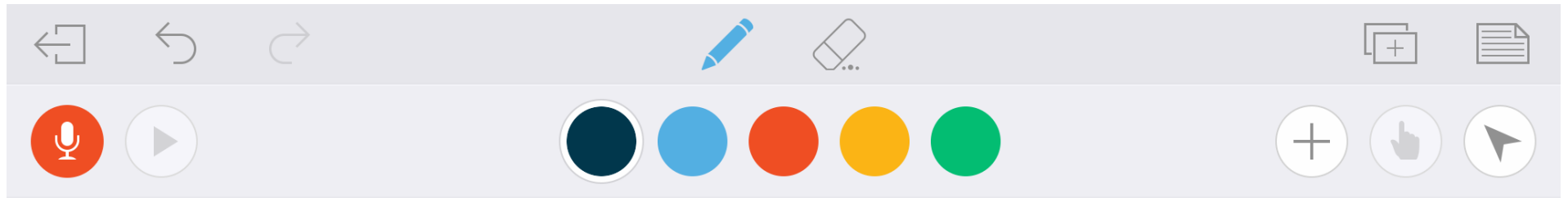

$$\Sigma F_x = m a_x$$

$$f_k = m a_x$$

$$\uparrow$$
$$\mu_k N = m a_x$$

$$\mu_k m g = m a_x$$

$$\boxed{a_x} = \mu_k g = (0.21)(9.8)$$

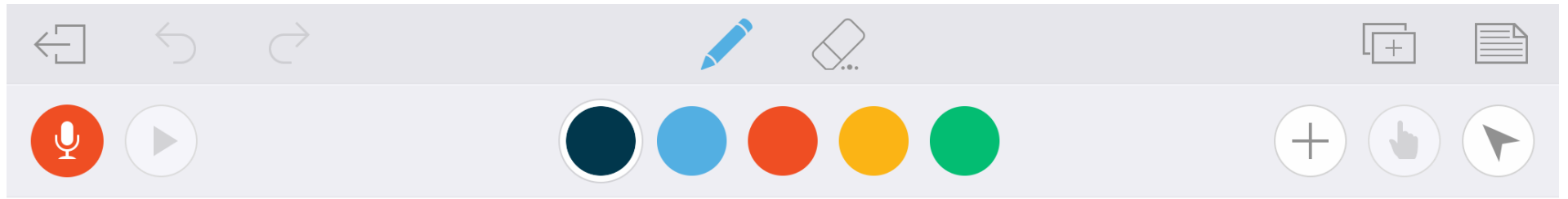


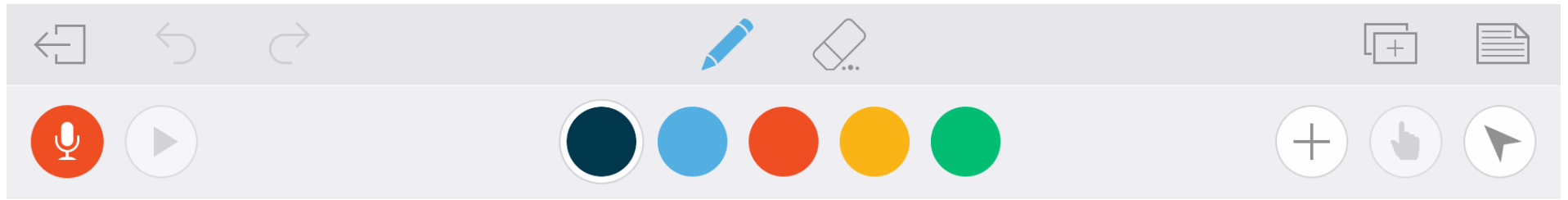
$$a_x = 0.21 \text{ g}$$

(t)

→ $a = \frac{V_f - V_i}{t} \rightarrow t = \frac{V_f - V_i}{a}$

$t = 2.58 \text{ s}$ $t = \frac{5.3}{(0.21)(\text{g})}$





$$\sum \vec{F}_x = m a_x$$

$$\underline{T - f_k - m_1 g \sin \theta}$$

$$= m_1 a_x$$

$$\sum \vec{F}_y = m a_y = 0$$

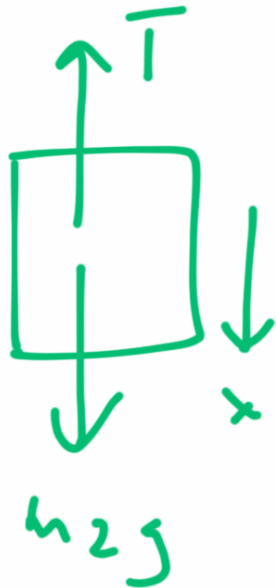
$$N - m_1 g \cos \theta = 0$$

$$N = m_1 g \cos \theta$$

$$f_k = \mu_k m_1 g \cos \theta$$

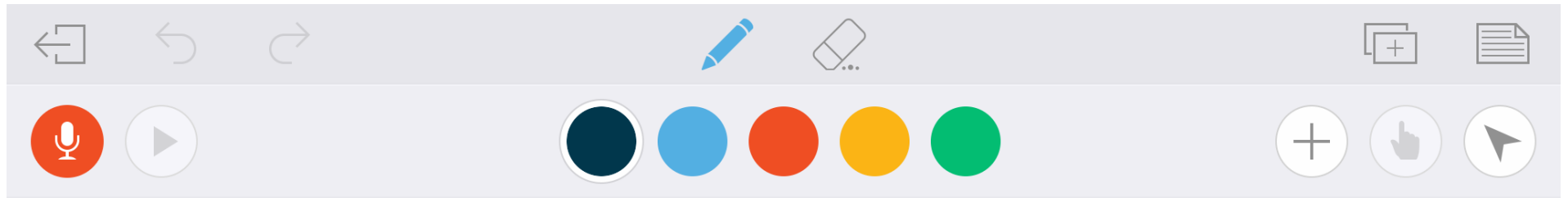
$$\boxed{T - \mu_k m_1 g \cos \theta - m_1 g \sin \theta = m_1 a_x}$$

$$\textcircled{T} - \mu_k m_1 g \cos \theta - m_1 g \sin \theta = m_1 a_x$$

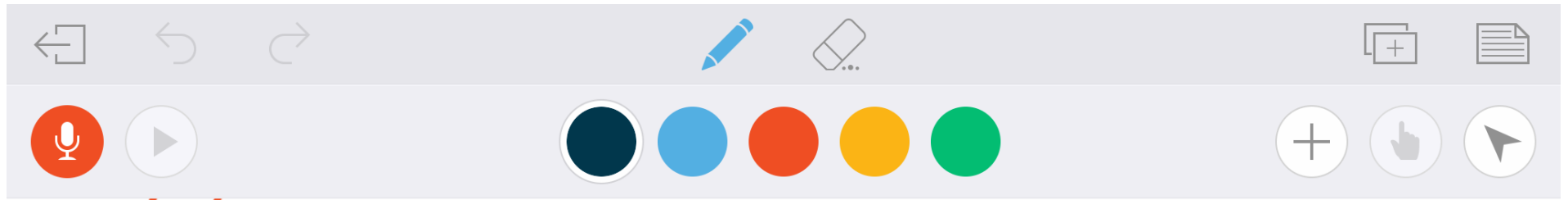


$$m_2 g - \textcircled{T} = m_2 a_x$$

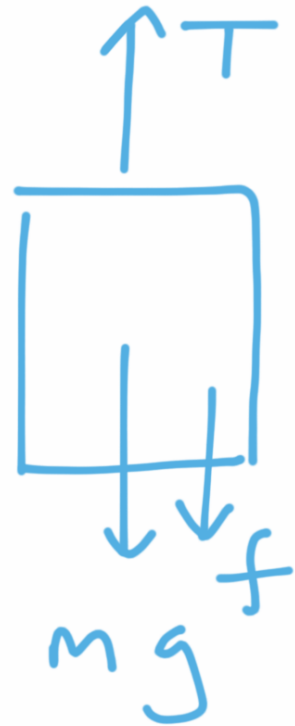
$$m_2 g - \mu_k m_1 g \cos \theta - m_1 g \sin \theta = (m_1 + m_2) a_x$$



$$a_x = \left(\frac{m_2 - m_1 \sin \theta}{m_1 + m_2} \right) g$$
$$= 2.43 \text{ m/s}^2$$



8.



$$f \approx 140 \text{ N}$$

Work-Energy
Theorem

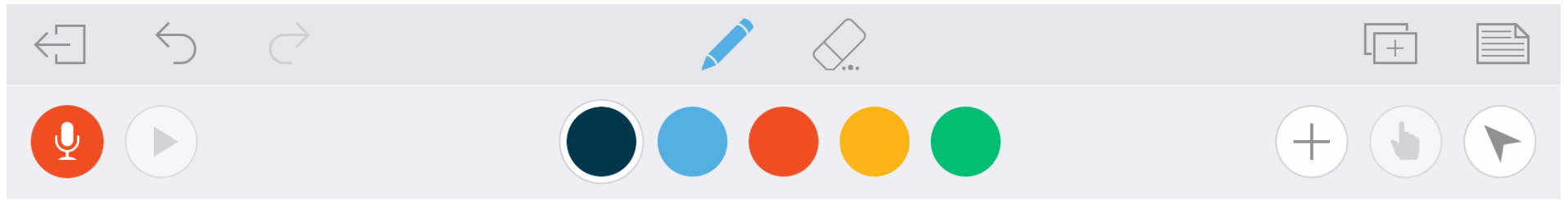
$$W_T +ve$$

$$W_{mg} = -ve$$

$$W_f = -ve$$

$$W_{\text{all forces}} = 0$$

$$\begin{aligned} W_{\text{all forces}} &= \Delta K \\ &= \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 \\ &= 0 \end{aligned}$$



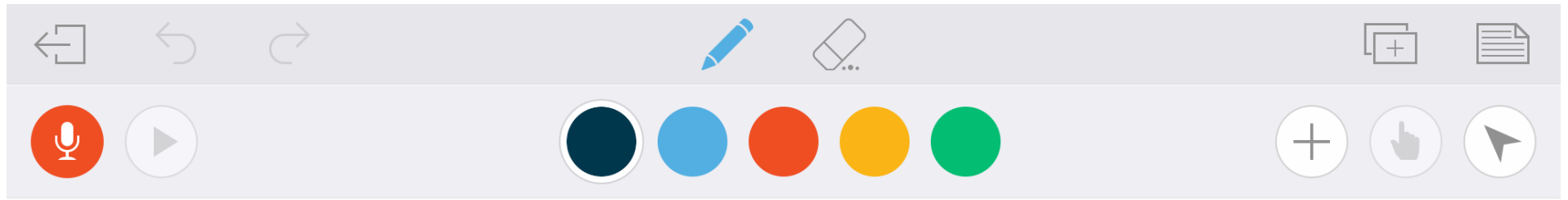
$$W_T + W_{ms} + W_f = 0$$

$$W_T + (-mgh) + (-fh) = 0$$

$$W_T = mgh + fh$$

$$= 636,090 \text{ J}$$

$$(6.36 \times 10^5)$$



b) $W_{mg} = -mgh$
 $= -630,630 \text{ J}$

c) $W_f = -fh$
 $= -5460 \text{ J}$

d) ~~W~~ $W_{\text{man}} = 0$ d) W_T